

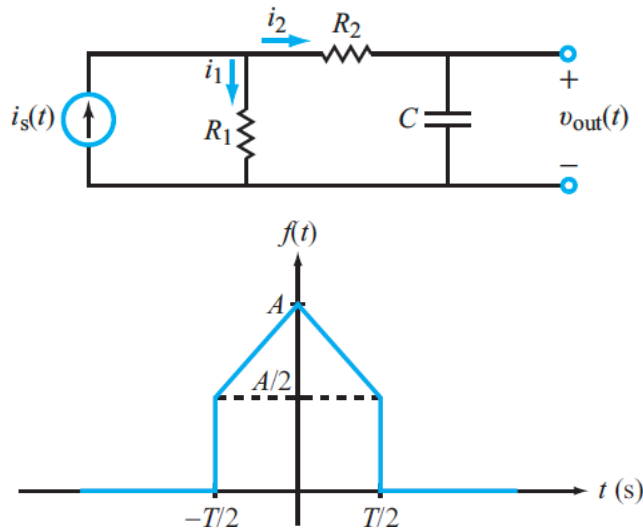
1. (30 pts) Given that the Fourier transform of the signal $x(t)$ is $X(\omega) = \frac{5j\omega}{2+j\omega}$, calculate the Fourier transforms of the following signals:

- (10 pts) $x(3t - 2)$
- (10 pts) $tx(t)$
- (10 pts) $\frac{dx(t)}{dt}$

2. (20 points) Given that the Fourier transform of the signal $x(t)$ is $X(\omega) = \frac{j\omega e^{-j\omega}}{2+j\omega} + 1$, calculate the Fourier transforms of the following signals:

- (10 pts) $x(\frac{5}{8}t)$
- (10 pts) $x(t)\cos(2t)$

3. (50 pts) Consider the following circuit, and the input waveform $i_s(t) = f(t)$:



- (25 pts) Derive an expression for $I_s(\omega)$ using Fourier analysis where $I_s(t) = f(t)$
- (25 pts) Calculate $V_{out}(\omega)$ for this circuit.